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RV COLLEGE OF ENGINEERING®

(An Autonomous Institution affiliated to VTU)

III Semester B. E. Regular / Supplementary Examinations Feb/Mar-2025

Industrial Engineering and Management

MANUFACTURING PROCESSES

Time: 03 Hours *Maximum Marks: 100*

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2 is compulsory. Answer any one full question from 3 and 4, 5 and 6, 7 and 8, 9 and 10.

PART-A

		M	BT	CO
1	Describe the importance of pattern color coding and provide examples.	2	1	1
2	What are chaplets and describe their purpose in casting.	2	1	2
3	Name the differences between hot working and cold working.	2	2	2
4	What is precision investment casting and list its advantages.	2	1	2
5	Define electron beam welding (EBW) and state where it is used.	2	1	1
6	List the factors influencing the weldability.	2	1	3
7	Write the different types of chips formed in metal cutting.	2	1	2
8	What is machinability and list the factors that affect it.	2	1	3
9	Find the machining time required for drilling a $25mm$ hole through a $50mm$ thick plate at $600rpm$ with a feed rate of $0.2mm/rev$.	2	1	3
10	Write about the purpose of indexing in milling.	2	1	4

PART-B

a	Discuss recent developments in manufacturing.	08	2	1
b	Illustrate the steps involved in sand moulding process with neat sketch.	08	3	2
a	Explain the process of forging and compare smith forging, drop forging and press forging process.	08	4	2
b	Explain the principle of extrusion, discuss its types according to direction of flow of materials and illustrate with a neat sketch.	08	4	2
	OR			
a	Discuss the principle of shell moulding process with neat sketch.	08	2	2
b	Explain the hot chamber die casting process with neat sketch.	08	4	2
a	Explain the principles of arc welding with basic welding circuit and its applications.	08	4	1
b	Identify the welding process that uses heat generated from burning a combination of different gases to melt and fuse metals and how does it work? Explain with neat sketch.	08	4	1

		OR	
6	a	Distinguish between fusion welding and solid state welding with examples.	
	b	Discuss the factors affecting weld quality and explain the causes and remedies of common welding defects.	
7	a	Explain single point cutting tool nomenclature and its geometry with neat sketch.	
	b	In an orthogonal cutting test with a tool of rake angle 10^0, the following observations were made: Chip thickness ratio = 0.3 Cutting force = $1290N$ Feed force = $1650N$ From Merchant's theory, calculate shear plane angle, coefficient of friction, velocity of chip relative to tool at the chip tool interface.	
		OR	
8	a	Explain the desired properties and types of cutting tool materials.	
	b	Discuss the different types of cutting fluids and their application.	
9	a	With a neat sketch, explain the nomenclature of twist drill bit.	
	b	Explain the different types of drilling related operations with neat sketch.	
		OR	
10	a	With neat sketch, differentiate between up milling and down milling operation.	
	b	A gear blank on the milling machine in order to cut 73 teeth, differential indexing is used to rotate the part. The machine has a 40:1 ratio index head. Calculate the gear ratio necessary between the spindle and index plate.	

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RV COLLEGE OF ENGINEERING®

(An Autonomous Institution Affiliated to VTU)

III Semester B. E. Supplementary Examinations June/July-2025

Industrial Engineering and Management

MANUFACTURING PROCESS

03 Hours *Maximum Marks: 100*

ctions to candidates:

Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.

Answer FIVE full questions from Part B. In Part B question number 2 is compulsory.

Answer any one full question from 3 and 4, 5 and 6, 7 and 8, 9 and 10.

PART-A

		M	BT	CO
1.1	Differentiate between job and batch production.	02	1	1
1.2	Why is casting considered suitable for complex shapes?	02	1	1
1.3	List the two common applications of rolling.	02	2	2
1.4	What is the principle behind centrifugal casting?	02	2	2
1.5	When would you choose adhesive bonding over welding?	02	3	3
1.6	Suggest ways to reduce welding defects in MIG welding.	02	3	3
1.7	What type of machining would you use for a flat surface with key slots?	02	2	2
1.8	Why is excessive cutting temperature harmful?	02	2	2
1.9	Why are advanced machining processes needed in modern manufacturing?	02	2	2
1.10	Suggest a product where LBM is suitable.	02	2	2

PART-B

a	Discuss the recent developments in manufacturing and their significance.	08	1	1
b	List the advantages and limitations of Expendable – Mould Casting Processes.	08	2	2
a	Compare hot working and cold working process.	08	2	2
b	What is press forging? How does it differ from drop forging?	08	2	2
	OR			
a	Describe the principle of Investment Casting and list its application.	08	3	3
b	Explain the steps of centrifugal casting process with a neat sketch.	08	3	3
a	Which fusion-welding process would you select for welding thick steel plates, and why?	08	3	3
b	Evaluate how poor weldability can affect the quality and performance of a welded structure.	08	3	3
	OR			
	Explain the process of soldering and identify its common areas of application.	08	4	4
	List and discuss welding defects and with their causes.	08	4	4

7	a	Explain the role of built-up edge in the cutting process and how it affects machining performance.	
	b	State assumptions of merchant circle diagram with a neat sketch.	
		OR	
8	a	List and describe the various tool angles used in single-point cutting tools.	
	b	Explain the significance of tool life in metal cutting operations and its effect on production cost and quality.	
9	a	Which advanced machining process would you recommend for machining intricate holes in hard metals without tool wear? Justify your choice.	
	b	Explain the working principle of Electrochemical Machining (ECM). How is it different from conventional?	
		OR	
10	a	Discuss the limitations of Abrasive-Jet Machining (AJM) for precision machining applications.	
	b	Explain the concept of hybrid machining. Why is it used instead of conventional or single advanced processes?	

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RV COLLEGE OF ENGINEERING®

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III Semester B. E. Examinations April/May-2024

Industrial Engineering and Management

MANUFACTURING PROCESSES

e: 03 Hours *Maximum Marks: 100*

uctions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2 is compulsory. Answer any one full question from 3 and 4, 5 and 6, 7 and 8, 9 and 10.

PART-A

		M	BT	CO
1	The value of chip reduction coefficient depends upon _____ cutting tool material and _____.	2	1	1
2	In the Taylor's tool life equation, $VTn = C$, the value of $n = 0.5$. The tool life has $180 minutes$ at a cutting speed of $18 m/min$. If the tool life is reduced to $45 minutes$, then the cutting speed will be _____.	2	3	3
3	The working of tin and lead even at room temperature is termed as hot working. Give reasons.	2	3	2
4	List any two defects of welding.	2	2	2
5	State any two differences between orthogonal cutting and oblique cutting.	2	2	1
6	Define porosity.	2	2	3
7	List any two limitations of extrusion process.	2	1	2
8	List any two applications of hydro forming process.	2	3	1
9	Circumference of a work piece is to be divided into 60 equal divisions using simple indexing method having 40 worm wheel teeth. Calculate indexing movement.	2	3	2
10	List the cutting parameters that greatly affects the tool life.	2	2	2

PART-B

a	Define pattern. List the types of patterns and explain any one briefly.	06	2	2
b	Explain the process of casting with a neat sketch.	10	1	1
a	With a neat sketch, explain the centrifugal casting process which is best suited for casting of hollow pipes and tubes.	10	3	3
b	List the advantages and disadvantages of extrusion process.	06	2	2
	OR			
a	How are cup shaped vessels being manufactured? Explain the metal spinning process with a neat sketch.	10	3	3
b	List the advantages and disadvantages of hydro forming process.	06	2	2

5	a	With a neat sketch, explain resistance seam welding process which is best used to rectangular or square or even in circular shapes.	
	b	State the differences between fusion and solid state welding process.	
		OR	
6	a	Explain with the help of sketch, working principle of Thermit Welding.	
	b	What is weld quality? Explain the factors that affect weldability.	
7	a	List cutting tool materials. Describe in brief, stating its principal characteristics and applications of any two cutting tool materials.	
	b	Derive an expression for the shear angle in orthogonal cutting, in terms of rake angle and chip thickness ratio.	
		OR	
8	a	What are the different types of chips that are formed in metal matching?	
	b	An orthogonal cutting operation is performed using a rake angle of 15^0, chip thickness before the cut $= 0.012inch$ and width of cut $= 0.100inch$. The chip thickness ratio is measured after the cut to be 0.55. Determine i) The chip thickness after the cut ii) Shear angle iii) Friction angle iv) Coefficient of friction v) Shear strain.	
9	a	Define indexing. Explain the following indexing methods. i) Direct indexing ii) Compound indexing.	
	b	A drilling operation is performed on a workpiece made of mild steel using a twist drill with a diameter of $10mm$. The cutting speed is set to $60m/min$ and the feed rate is $0.1mm/rev$. The drill needs to penetrate the workpiece to a depth of $30mm$. Assume that the drill needs to be changed after every 50 holes and each tool change takes $2minutes$. Calculate the total machining time required to drill 200 holes in the workpiece.	
		OR	
10	a	State different types of drilling operations and explain any two with a neat sketch.	
	b	Differentiate between up milling and down milling.	